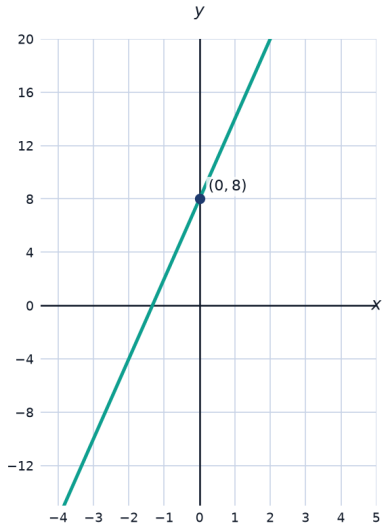
Similarity & Proportional Reasoning

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

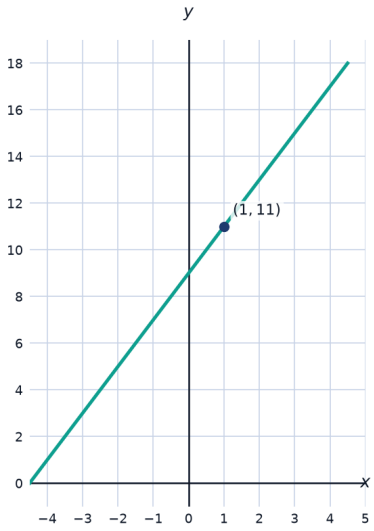
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 8$ in rotations, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 2x + 9$ in dilations, evaluate $f(2)$.



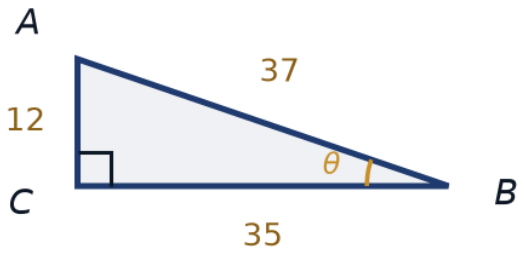
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for dilations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

On the enhanced ACT Math test (45 questions in 50 minutes), a student allocates 10 minutes to the first 15 questions and 16 minutes to questions 16 through 30. How many seconds per question remain for the final 15 questions, assuming 3 minutes are reserved at the end to check answers?

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 1 · FoundationsThe Digital SAT Math section allocates 70 minutes across two modules of equal length. If each module contains 22 questions, what is the exact average time available per question in seconds?

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the least common multiple of 6 and 8.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 4 · StrategyA student finishes the first 18 questions of Module 1 with 9 minutes remaining. Questions 19 and 20 are lengthy geometry proofs, while Questions 21 and 22 are quick Desmos system intersection problems. In what order should the student attack the remaining 4 questions, and why?

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 2 · MediumA student solves a linear equation and obtains the fraction $x = -\frac{7}{4}$. What are the valid ways to enter this answer into the Digital SAT Student-Produced Response (SPR) grid?

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Practice 1.2: If a and b are positive integers such that the greatest common factor of a and b is 6, and the least common multiple of a and b is 72, what is the product $a \times b$?

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11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Practice 2.2: What is the sum of the solutions of the quadratic equation $3x^2 - 12x - 5 = 0$?

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 2 · MediumA study of 100 students recorded test results based on exam prep: • Studied: 54 passed, 6 failed (60 total) • Did not study: 18 passed, 22 failed (40 total) What is the probability that a randomly chosen student passed the exam, given that the student studied?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Why should you test Choice C (or H) first when back-solving numerical answer choices on the ACT?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the greatest common factor (GCF) and least common multiple (LCM) of 24 and 36?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Transformations & Symmetry

Practice 2.1: For what values of x is $|2x - 3| < 7$?

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Foundations & Reasoning

What is the sum and product of the roots of $3x^2 - 15x + 7 = 0$?

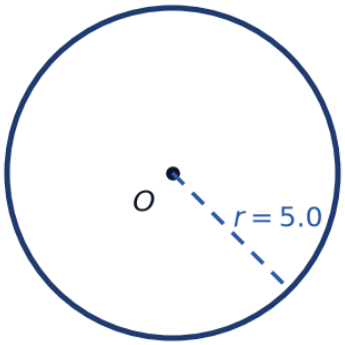
17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 10.01 Coordinate Geometry

For what values of x does $|2x - 5| > 9$ hold?

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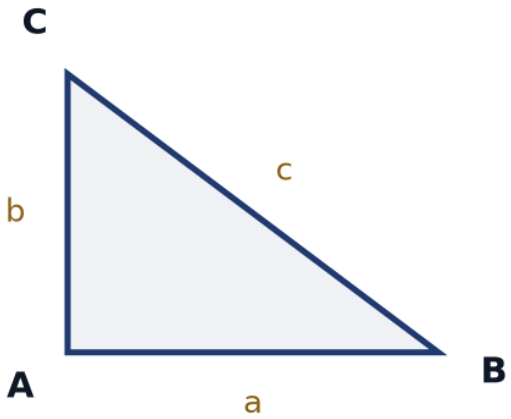
18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Probability

The equation of a circle in the xy-plane is $(x-5)^2+(y+2)^2=81$. What is the radius of the circle?



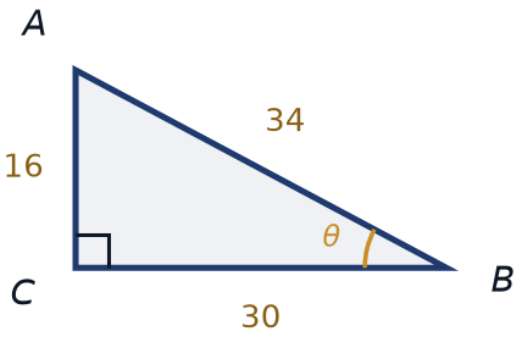
19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Lines & Angles

A right triangle has legs 9 and 12. Find the hypotenuse.



20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Triangles & Congruence

A triangle has angles 50° and 60°. Find the third.

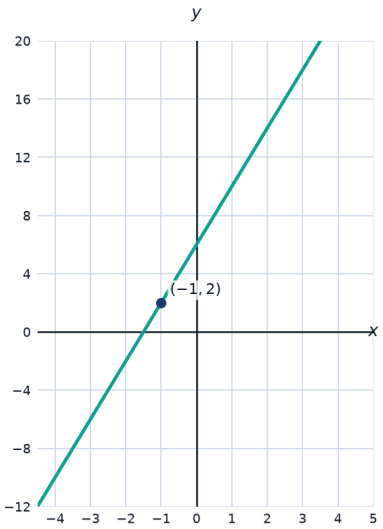


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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 6$ in translations, evaluate $f(2)$.



a. 14

b. 18

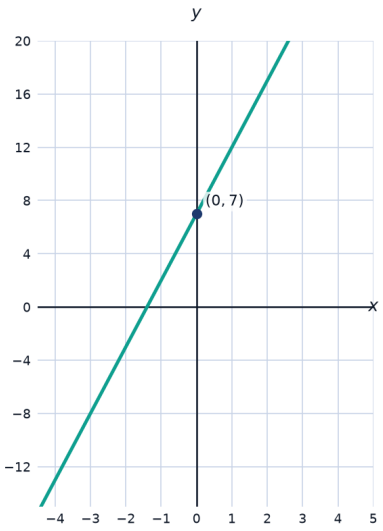
c. 8

d. 18

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 7$ in reflections, evaluate $f(2)$.



a. 17

b. 22

c. 10

d. 21

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for translations consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for rotations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 4x + 4$ in compositions of transformations, evaluate $f(3)$.

a. 16

b. 20

c. 12

d. 20

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In the figure, right triangle ABC has a right angle at B, and altitude $\overline{BD}$ is drawn to hypotenuse $\overline{AC}$. The altitude divides the hypotenuse into segments of lengths $AD = 9$ and $DC = 16$. Two similar right-triangular prisms, $\mathcal{P}_1$ and $\mathcal{P}_2$, have bases ΔABD and ΔBCD, respectively. If the volume of prism $\mathcal{P}_1$ is $135\ \text{cm}^3$, what is the volume of prism $\mathcal{P}_2$, in cm^3?

a. 320

b. 240

c. 625

d. 180

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the diagram below, line segment DE is parallel to line segment BC. If $AD = 6$ cm, $DB = 4$ cm, and $AE = 9$ cm, what is the length of EC?

a. 6 cm

b. 15 cm

c. 13.5 cm

d. 54 cm

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the figure shown, line segments AC and BD intersect at point E, and $\overline{AB} \parallel \overline{CD}$. If $AB = 9$, $CD = 15$, and $AE = 6$, what is the length of segment EC?

a. 3.6

b. 9

c. 10

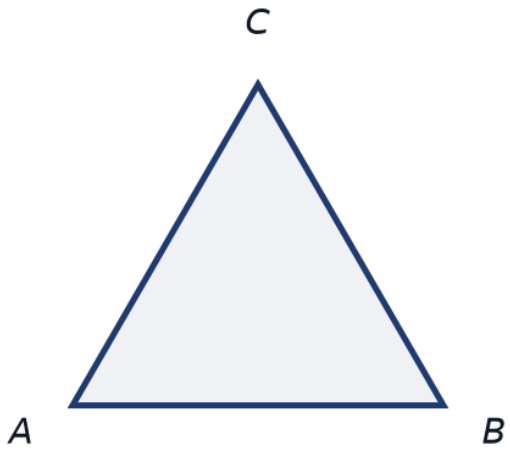
d. 12

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

What is the converse of the statement: "If A triangle is equilateral, then it is equiangular."?



a. If A triangle is not equilateral, then it is not equiangular.

b. If It is not equiangular, then a triangle is not equilateral.

c. If It is equiangular, then a triangle is equilateral.

d. If It is equiangular, then a triangle is not equilateral.

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Consider the following conditional statement where the universe of discourse is the set of all parallelograms: x Which of the following correctly classifies the truth values of Statement P, its converse, its inverse, and its contrapositive, along with a mathematically valid counterexample for any false statement(s)?

a. Statement P is False (counterexample: a rhombus with unequal diagonal lengths); Converse is True; Inverse is True; Contrapositive is False (counterexample: a rhombus with unequal diagonal lengths).

b. Statement P is False (counterexample: a rectangle with unequal adjacent sides); Converse is True; Inverse is True; Contrapositive is False (counterexample: a rectangle with unequal adjacent sides).

c. Statement P is False (counterexample: a rhombus with unequal diagonal lengths); Converse is False (counterexample: a non-square rhombus); Inverse is False (counterexample: a non-square rectangle); Contrapositive is False (counterexample: a rhombus with unequal diagonal lengths).

d. Statement P is True; Converse is False (counterexample: a rhombus with unequal diagonal lengths); Inverse is False (counterexample: a rhombus with unequal diagonal lengths); Contrapositive is True.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Point P lies on the directed line segment AB such that AP:PB = 2:3. Given the coordinates of A as $(-2, 5)$ and the coordinates of B as $(8, -5)$, find the coordinates of point P.

a. $(2, 2)$

b. $(4, -1)$

c. $(0, 0)$

d. $(6, -3)$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the statement: "If a quadrilateral has four right angles, then it is a square." ; ; Which of the following represents the inverse of this statement, and what is its truth value? ; ; Statement P: A quadrilateral has four right angles. ; Statement Q: It is a square.

a. If a quadrilateral is not a square, then it does not have four right angles. (True)

b. If a quadrilateral does not have four right angles, then it is not a square. (True)

c. If a quadrilateral is not a square, then it does not have four right angles. (False)

d. If a quadrilateral does not have four right angles, then it is not a square. (False)

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Point M is the midpoint of segment $\overline{AB}$. If $AM = 3x + 1$ and $MB = 5x - 9$, what is the length of segment $\overline{AB}$?

a. 5

b. 16

c. 26

d. 32

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Two vertical poles of heights 10 m and 15 m stand on level ground. Guy wires connect the top of each pole to the base of the other pole. At what height above the ground do the two crossing wires intersect?

a. 7.5 m

b. 5 m

c. 6 m

d. 6.25 m

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

To determine the height of a sea cliff, a surveyor places a small mirror flat on horizontal ground 18 meters from the base of the cliff. The surveyor steps back 2 meters from the mirror until the top of the cliff is aligned with the center of the mirror. If the surveyor's eye level is 1.6 meters above the ground, what is the height of the cliff?

a. 16.2 m

b. 22.5 m

c. 14.4 m

d. 18.0 m

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A pinhole camera has its pinhole at point E. An object AB of height 15 cm is placed 45 cm in front of E. The inverted image CD forms on a screen 30 cm behind E. Assuming rays travel in straight lines intersecting at E parallel to the camera plates, what is the image height CD?

a. 10 cm

b. 12 cm

c. 8 cm

d. 7.5 cm

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In ΔABC, segments $DE \parallel FG \parallel BC$ with D, F on AB and E, G on AC. If $AD = 2$, $DF = 3$, $FB = 5$, and base $BC = 30$, what is the difference between the lengths of segments FG and DE?

a. 6

b. 9

c. 12

d. 15

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Probability

A 6-foot-tall person stands 10 feet away from the base of a 16-foot streetlight. How long is the person's shadow cast along the ground?

a. 4 ft

b. 8 ft

c. 5 ft

d. 6 ft

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Lines & Angles

From an exterior point P, two secant lines are drawn to a circle. The first secant intersects the circle at A and B with $PA = 4$ and $AB = 8$. The second secant intersects the circle at C and D with $PC = 3$. What is the length of chord CD?

a. 16

b. 12

c. 13

d. 15

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Triangles & Congruence

Two isosceles triangles each have a vertex angle measuring 40°. The leg of the smaller triangle has length 15, and the corresponding leg of the larger triangle has length 25. If the base of the smaller triangle is 12, what is the base of the larger triangle?

a. 18

b. 24

c. 20

d. 22

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

The sides of ΔABC measure 7, 10, and 12. A similar triangle ΔDEF has a shortest side of length 21. What is the perimeter of ΔDEF?

a. 58

b. 87

c. 72

d. 96

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

In ΔABC, $\angle A = 42^\circ$ and $\angle B = 68^\circ$. In ΔDEF, $\angle D = 70^\circ$ and $\angle E = 68^\circ$. Which statement correctly determines whether the triangles are similar?

a. The triangles are similar by AA because $\angle C = 70^\circ$, matching $\angle D$

b. The triangles are not similar because $\angle A \neq \angle D$

c. The triangles are similar by SAS using sides adjacent to 68°

d. Similarity cannot be determined without side lengths

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Circles

Chords AB and CD intersect inside a circle at point P. If $PA = 4$, $PB = 9$, and $PC = 6$, what is the length of segment PD?

a. 6

b. 8

c. 5.4

d. 7

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Area, Surface Area & Volume

Which of the following conditions is NOT sufficient to prove that two triangles are similar?

a. Two pairs of corresponding angles are congruent (AA)

b. Three pairs of corresponding sides are proportional (SSS)

c. Two pairs of sides are proportional and a non-included angle is congruent (SSA)

d. Two pairs of sides are proportional and their included angle is congruent (SAS)

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Transformations & Symmetry

A cone with height 30 cm is sliced by a plane parallel to its base at a distance of 15 cm from the apex, creating a smaller top cone and a bottom frustum. What fraction of the original cone's volume is contained in the frustum?

a. 1/2

b. 3/4

c. 1/8

d. 7/8

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